

HW # 05 Solutions

Problem 1:

$$(a) \quad \langle x, y \rangle = x^\top \bar{y} = \sum_{i=1}^n x_i \bar{y}_i .$$

$$(i) \quad \overline{\langle y, x \rangle} = \overline{y^\top \bar{x}} = \overline{\sum_{i=1}^n y_i \bar{x}_i} = \sum_{i=1}^n \bar{y}_i x_i \\ = \langle x, y \rangle .$$

$$(ii) \quad \langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = (\alpha_1 x_1 + \alpha_2 x_2)^\top \bar{y} \\ = \alpha_1 x_1^\top \bar{y} + \alpha_2 x_2^\top \bar{y} \\ = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle .$$

$$(iii) \quad \langle x, x \rangle = \sum_{i=1}^n x_i \bar{x}_i = \sum_{i=1}^n |x_i|^2 .$$

Hence, $\langle x, x \rangle \geq 0$ for any $x \in \mathbb{C}^n$, and $\langle x, x \rangle = 0 \iff |x_i|^2 = 0, i = 1, \dots, n, \iff x = 0$.

$$(b) \quad \langle x, y \rangle = \bar{x}^\top y = \sum_{i=1}^n \bar{x}_i y_i .$$

$$(i) \quad \overline{\langle y, x \rangle} = \overline{\bar{y}^\top x} = \overline{\sum_{i=1}^n \bar{y}_i x_i} \\ = \sum_{i=1}^n y_i \bar{x}_i = \bar{x}^\top y = \langle x, y \rangle .$$

$$(ii) \quad \langle x, \beta_1 y_1 + \beta_2 y_2 \rangle = \bar{x}^\top (\beta_1 y_1 + \beta_2 y_2) \\ = \beta_1 \bar{x}^\top y_1 + \beta_2 \bar{x}^\top y_2 \\ = \beta_1 \langle x, y_1 \rangle + \beta_2 \langle x, y_2 \rangle .$$

$$(iii) \quad \langle x, x \rangle = \bar{x}^\top x = \sum_{i=1}^n \bar{x}_i x_i = \sum_{i=1}^n |x_i|^2 .$$

Hence, $\langle x, x \rangle \geq 0$ for any $x \in \mathbb{C}^n$, and $\langle x, x \rangle = 0 \iff |x_i|^2 = 0, i = 1, \dots, n, \iff x = 0$

Problem 2:

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$$

$$p_0(x) = 1, \quad p_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$p_1(x) = x, \quad p_4(x) = \frac{1}{2}(5x^3 - 3x).$$

$$\begin{aligned}
\langle \mathbf{p}_0, \mathbf{p}_3 \rangle &= \int_{-1}^1 \mathbf{p}_0(x) \mathbf{p}_3(x) dx \\
&= \int_{-1}^1 \frac{1}{2} (5x^3 - 3x) dx \\
&= \frac{1}{2} \left[\frac{5x^4}{4} - \frac{3x^2}{2} \right]_{-1}^1 \\
&= \frac{1}{2} \left[\left(\frac{5}{4} - \frac{3}{2} \right) - \left(\frac{5}{4} - \frac{3}{2} \right) \right] \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
\langle \mathbf{p}_1, \mathbf{p}_2 \rangle &= \int_{-1}^1 \mathbf{p}_1(x) \mathbf{p}_2(x) dx \\
&= \int_{-1}^1 x \frac{1}{2} (3x^2 - 1) dx \\
&= \int_{-1}^1 \frac{1}{2} (3x^3 - x) dx \\
&= \frac{1}{2} \left[\frac{3x^4}{4} - \frac{x^2}{2} \right]_{-1}^1 \\
&= \frac{1}{2} \left[\left(\frac{3}{4} - \frac{1}{2} \right) - \left(\frac{3}{4} - \frac{1}{2} \right) \right] \\
&= 0.
\end{aligned}$$

Problem 3:

$$v^1 = y^1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \|v^1\|^2 = 6.$$

$$v^2 = y^2 - a_{21}v^1.$$

$$\begin{aligned}
a_{21} &= \frac{\langle y^2, v^1 \rangle}{\|v^1\|^2} = \frac{\begin{bmatrix} 4 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}{6}, \\
&= \frac{3}{6} = \frac{1}{2}.
\end{aligned}$$

$$\therefore v^2 = \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3\frac{1}{2} \\ 1 \\ -1\frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix}.$$

$$\|v^2\|^2 = \frac{1}{4}(49 + 4 + 9) = \frac{62}{4} = \frac{31}{2}.$$

$$v^3 = y^3 - a_{31}v^1 - a_{32}v^2.$$

$$\begin{aligned}
a_{31} &= \frac{\langle y^3, v^1 \rangle}{\|v^1\|^2} = \frac{\begin{bmatrix} -2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}{6}, \\
&= -\frac{3}{6} = -\frac{1}{2}. \\
a_{32} &= \frac{\langle y^3, v^2 \rangle}{\|v^2\|^2} = \frac{\begin{bmatrix} -2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix} \left(\frac{1}{2}\right)}{3\frac{1}{2}}, \\
&= -\frac{19}{31}. \\
v^3 &= \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \frac{19}{31} \begin{bmatrix} 3\frac{1}{2} \\ 1 \\ -1\frac{1}{2} \end{bmatrix} \\
&= \begin{bmatrix} 40 \\ 100 \\ 160 \end{bmatrix} \left(\frac{1}{62}\right) \approx \begin{bmatrix} 0.65 \\ 1.61 \\ 2.58 \end{bmatrix}.
\end{aligned}$$

Problem 4:

- (a) If A^{-1} , C^{-1} and $(C^{-1} + DA^{-1}B)^{-1}$ each exist, then

$$M = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1} \quad (*)$$

is well defined (means each of the terms exists and the indicated matrix products are compatible).

To show: The proposed inverse $(*)$ when multiplied by $(A + BCD)$ gives the identity.

We will show $M(A + BCD) = I$.

$$\begin{aligned}
M(A + BCD) &= I - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}D + A^{-1}BCD - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}BCD \\
&= I - A^{-1}B[(C^{-1} + DA^{-1}B)^{-1}C^{-1} - I + (C^{-1} + DA^{-1}B)^{-1}DA^{-1}B]CD \\
&= I - A^{-1}B[(C^{-1} + DA^{-1}B)^{-1}(C^{-1} + DA^{-1}B) - I]CD \\
&= I - A^{-1}B[I - I]D \\
&= I.
\end{aligned}$$

- (b)

$$A^{-1} = \text{diag}([1, 2, 2, 1, 2]),$$

$$C^{-1} = 5,$$

$$A^{-1}B = \begin{bmatrix} 1 \\ 0 \\ 4 \\ 0 \\ 6 \end{bmatrix},$$

$$DA^{-1}B = \begin{bmatrix} 1 & 0 & 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 4 \\ 0 \\ 6 \end{bmatrix} = 27,$$

$$C^{-1} + DA^{-1}B = 32 \implies (C^{-1} + DA^{-1}B)^{-1} = \frac{1}{32}.$$

$$DA^{-1} = \begin{bmatrix} 1 & 0 & 4 & 0 & 6 \end{bmatrix},$$

$$\begin{aligned} \therefore (A + BCD)^{-1} &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 4 \\ 0 \\ 6 \end{bmatrix} \left(\frac{1}{32} \right) \begin{bmatrix} 1 & 0 & 4 & 0 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} - \frac{1}{32} \begin{bmatrix} 1 & 0 & 4 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 16 & 0 & 24 \\ 0 & 0 & 0 & 0 & 0 \\ 6 & 0 & 24 & 0 & 36 \end{bmatrix} \\ &= \frac{1}{32} \begin{bmatrix} 31 & 0 & -4 & 0 & -6 \\ 0 & 64 & 0 & 0 & 0 \\ -4 & 0 & 48 & 0 & -24 \\ 0 & 0 & 0 & 32 & 0 \\ -6 & 0 & -24 & 0 & 28 \end{bmatrix}. \end{aligned}$$

Problem 5:

(a) The naive estimate is plotted in Fig. 1

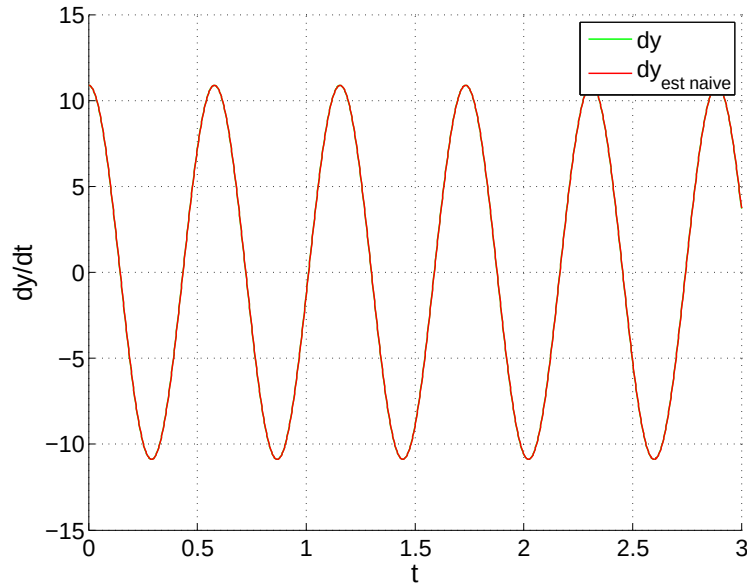


Figure 1: Naive Estimate

(b) We define Y_k so that it contains $M \geq 2$ of the “most recent” data points

$$Y_k = \begin{bmatrix} y[k - M + 1] \\ \vdots \\ y[k] \end{bmatrix},$$

where $y[k] = y(k\Delta T)$. For basis functions, we take the monomials, but you can use any set of independent functions for which you can compute the derivative. We let $\varphi_i(t) = t^i$, where $\varphi_0(t) = 1$.

Suppose that at time $t_k = k\Delta T$, we regress the data against $\{\varphi_0(t), \dots, \varphi_N(t)\}$, in other words,

$$y(t) = \alpha_0 + \alpha_1 t + \dots + \alpha_N t^N.$$

We then have

$$Y_k = A_k \alpha$$

where

$$A_k = \begin{bmatrix} 1 & (k - M + 1)\Delta T & \dots & ((k - M + 1)\Delta T)^N \\ 1 & (k - M + 2)\Delta T & \dots & ((k - M + 2)\Delta T)^N \\ \vdots & \vdots & \dots & \vdots \\ 1 & (k - 2)\Delta T & \dots & ((k - 2)\Delta T)^N \\ 1 & (k - 1)\Delta T & \dots & ((k - 1)\Delta T)^N \\ 1 & k\Delta T & \dots & (k\Delta T)^N \end{bmatrix}$$

which depends on k , and thus changes step-to-step. We need $M \geq N + 1$ for the columns of the matrix to be linearly independent. At the k -th step we have

$$\alpha = (A_k^\top A_k)^{-1} A_k^\top Y_k$$

We plug these coefficients back into

$$y(t) = \alpha_0 + \alpha_1(t) + \dots + \alpha_N(t)^N,$$

we differentiate it, evaluate it at whatever time we desire, and use that for our estimate of $\dot{y}(t)$. This is an acceptable solution, but a much more practical solution is available to us.

Suppose instead that at time t_k , we regress the data against $\{\varphi_0(t - t_k), \dots, \varphi_N(t - t_k)\}$, in other words,

$$y(t) = \alpha_0 + \alpha_1(t - t_k) + \dots + \alpha_N(t - t_k)^N.$$

All we are doing is shifting the time origin to t_k . By doing this, we end up with

$$Y_k = A \alpha$$

where

$$A = \begin{bmatrix} 1 & (-M + 1)\Delta T & \dots & ((-M + 1)\Delta T)^N \\ 1 & (-M + 2)\Delta T & \dots & ((-M + 2)\Delta T)^N \\ \vdots & \vdots & \dots & \vdots \\ 1 & -2\Delta T & \dots & (-2\Delta T)^N \\ 1 & -\Delta T & \dots & (-\Delta T)^N \\ 1 & 0 & \dots & 0 \end{bmatrix}$$

which does not change from one time step to the next. We still need $M \geq N + 1$ for the columns of the matrix to be linearly independent. At the k -th step we have

$$\alpha = (A^\top A)^{-1} A^\top Y_k$$

and we only need to compute $(A^\top A)^{-1} A^\top$ once. This is what I do on my robots. The calculation of the inverse is done off-line and stored.

We now compute

$$\dot{y}(t) = \alpha_1 + 2\alpha_2(t - t_k) + \cdots + N\alpha_N(t - t_k)^{(N-1)},$$

and thus

$$\dot{y}(t) = \begin{bmatrix} 0, 1, 2(t - t_k), \dots, N(t - t_k)^{(N-1)} \end{bmatrix} \alpha$$

Setting $t = t_k$, we obtain

$$\hat{y}_k = [0, 1, 0, \dots, 0] \alpha,$$

in other words,

$$\hat{y}_k = RY_k$$

where

$$R = [0, 1, 0, \dots, 0] (A^\top A)^{-1} A^\top.$$

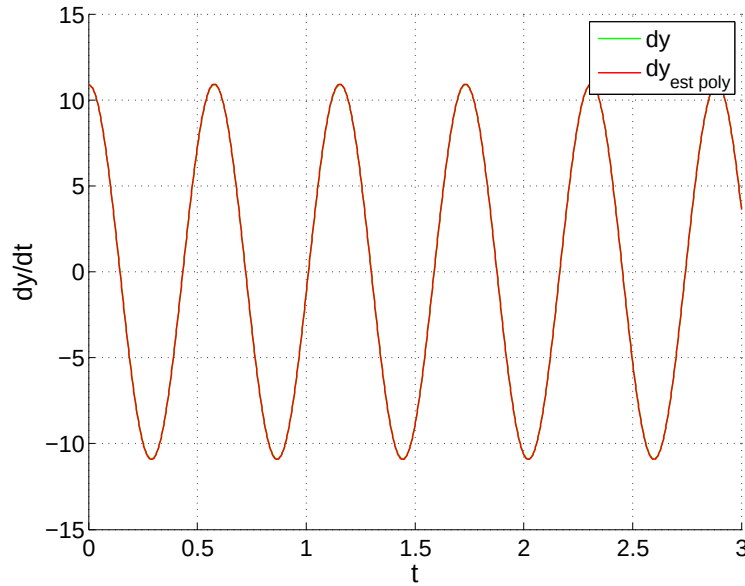


Figure 2: Regression

Choosing $M = 4$ and $N = 2$, we obtain the plot of the derivative given in Fig. 2. It looks exactly the same as the naive derivative, so we are disappointed that we worked so hard! **Remark: If you take $M = 2$ and $N = 1$ you get exactly the naive derivative.**

Problem 6:

- (a) The naive estimate is plotted in Fig. 3. Computing the error gives

$$\frac{\|\dot{y}_k - \hat{y}_k\|}{\text{Length of the data vector}} = 0.093,$$

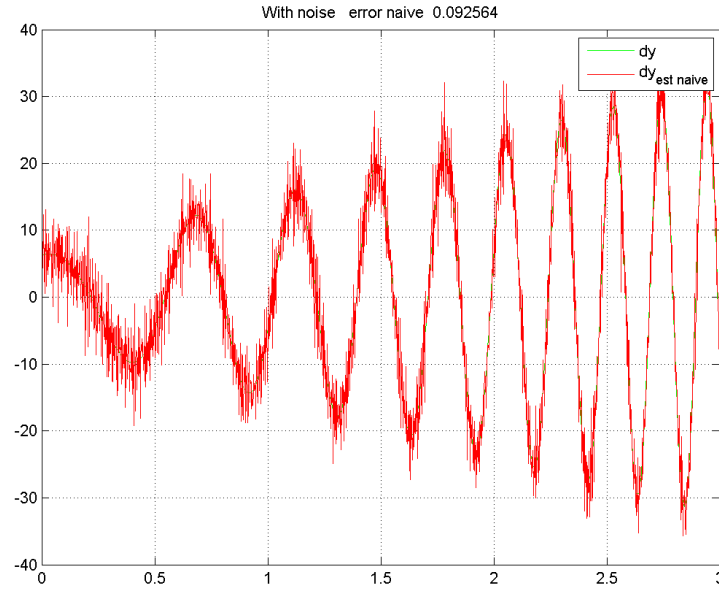


Figure 3: Naive Estimate

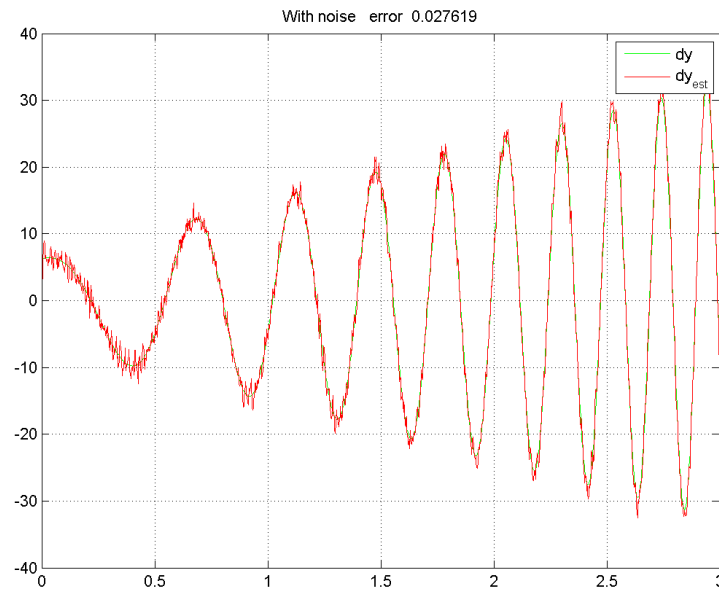


Figure 4: Regression

- (b) We do the same regression analysis as in Problem 5. We play around with the parameters a bit and settle on $M = 10$ and $N = 2$. We obtain

$$\frac{\|\dot{y}_k - \hat{y}_k\|}{\text{Length of the data vector}} = 0.027,$$

and a plot of the derivative is given in Fig. 4

Problem 7: We apply the normal equations:

$$\hat{x} = \alpha_1 y^1 + \alpha_2 y^2,$$

where $G^\top \alpha = b$ and

$$G = \begin{bmatrix} \langle y^1, y^1 \rangle & \langle y^2, y^1 \rangle \\ \langle y^1, y^2 \rangle & \langle y^2, y^2 \rangle \end{bmatrix},$$

$$b = \begin{bmatrix} \langle x, y^1 \rangle \\ \langle x, y^2 \rangle \end{bmatrix}.$$

Doing the calculations, we have

$$\langle y^1, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \right) = 5.$$

$$\langle y^1, y^2 \rangle = \langle y^2, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 3 & 3 \\ 0 & 0 \end{bmatrix} \right) = 3.$$

$$\langle y^2, y^2 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \right) = 4.$$

$$\langle x, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 4 & 0 \\ -1 & 0 \end{bmatrix} \right) = 4.$$

$$\langle x, y^2 \rangle = \text{tr} \left(\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \right) = 1.$$

$$\therefore \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

$$\therefore \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 13 \\ -7 \end{bmatrix} = \begin{bmatrix} 1.18 \\ -0.64 \end{bmatrix}.$$

$$\hat{x} = \frac{13}{11} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} - \frac{7}{11} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{6}{11} & -\frac{7}{11} \\ \frac{19}{11} & -\frac{7}{11} \end{bmatrix}.$$

Problem 8: Let $\gamma := d(x, M)$, and suppose that $m_1, m_2 \in M$ satisfy $\|x - m_i\| = \gamma$.

To Show $m_1 = m_2$ when the norm is strict.

Because M is a subspace, $\frac{m_1 + m_2}{2} \in M$.

Hence,

$$\begin{aligned} \gamma &= \inf_{y \in M} \|x - y\| \leq \left\| x - \frac{m_1 + m_2}{2} \right\| \\ &= \left\| \frac{x - m_1}{2} + \frac{x - m_2}{2} \right\| \\ &\leq \frac{1}{2} \|x - m_1\| + \frac{1}{2} \|x - m_2\| \\ &= \frac{\gamma}{2} + \frac{\gamma}{2} \\ &= \gamma. \end{aligned}$$

Hence, $\|(x - m_1) + (x - m_2)\| = \|x - m_1\| + \|x - m_2\|$.

Because the norm is strict, $\exists \alpha \geq 0$ such that either

(i) $(x - m_1) = \alpha(x - m_2)$ or

(ii) $(x - m_2) = \alpha(x - m_1)$.

In either case, we deduce from $\gamma = \|x - m_1\| = \|x - m_2\|$, that $\gamma = \alpha\gamma$, and, because $\gamma \neq 0$, we have $\alpha = 1$. With $\alpha = 1$, we have $x - m_1 = x - m_2$, and thus $m_1 = m_2$. $\square$

Problem 9:

(a) Not strictly normed. Let

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then $\|x + y\|_1 = 2 = \|x\|_1 + \|y\|_1$, but there does not exist any $\alpha \geq 0$ such that either $x = \alpha y$ or $y = \alpha x$.

(c) Not strictly normed. Let

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Then $6 = \|x + y\|_\infty = \|x\|_\infty + \|y\|_\infty$, but there does not exist any $\alpha \geq 0$ such that either $x = \alpha y$ or $y = \alpha x$.

(b) Strictly normed. The result is true for any norm induced by an inner product. Hence we give the proof for $\|x\| = \langle x, x \rangle^{1/2}$.

Let $x, y \in X$

Case 1 Either x or y is zero. Then $\|x + y\| = \|x\| + \|y\|$ is always true and either $x = 0 \cdot y$ or $y = 0 \cdot x$ holds. $\square$

Case 2 Both $x \neq 0$ and $y \neq 0$, but $\{x, y\}$ is linearly dependent. Then $x = \alpha y$ for some $\alpha \in \mathbb{R}$. It follows that $\|x + y\| = \|1 + \alpha\| \cdot \|y\|$ and $\|x\| + \|y\| = (1 + |\alpha|)\|y\|$. Because $y \neq 0$, we have

$$\|x + y\| = \|x\| + \|y\| \iff |1 + \alpha| = 1 + |\alpha| \iff \alpha \geq 0.$$

Hence, $\|x + y\| = \|x\| + \|y\| \iff x = \alpha y, \alpha \geq 0$. $\square$

Case 3 $\{x, y\}$ is linearly independent. By the Gram-Schmidt procedure, there exists $v \in X$ such that $x \perp v$ and $\text{span}\{x, y\} = \text{span}\{x, v\}$. Write $y = \alpha_1 x + \alpha_2 v$, so that

$$\begin{aligned} x + y &= (1 + \alpha_1)x + \alpha_2 v, \quad \|x + y\| = \|x\| + \|y\| \\ \text{if, and only if, } &\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\| \end{aligned}$$

By the Pythagorean Theorem,

$$\begin{aligned} \|x + y\|^2 &= \|(1 + \alpha_1)x + \alpha_2 v\|^2 \\ &= (1 + \alpha_1)^2 \|x\|^2 + (\alpha_2)^2 \|v\|^2 \\ &= [1 + 2\alpha_1 + (\alpha_1)^2] \|x\|^2 + (\alpha_2)^2 \|v\|^2. \end{aligned}$$

Furthermore

$$\|y\|^2 = (\alpha_1)^2 \|x\|^2 + (\alpha_2)^2 \|v\|^2.$$

Hence,

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\| \quad \text{if, and only if,} \quad 2\alpha_1 \|x\|^2 = 2\|x\| \cdot \|y\|.$$

Because $\|x\| \neq 0$ from $\{x, y\}$ linear independent, we have

$$\alpha_1\|x\| = \|y\|.$$

$\therefore \alpha_1 \geq 0$. Moreover,

$$(\alpha_1)^2\|x\|^2 = \|y\|^2 = (\alpha_1)^2\|x\|^2 + (\alpha_2)^2\|v\|^2 \text{ if, and only if, } \alpha_2 = 0.$$

Hence,

$$\|x + y\| = \|x\| + \|y\| \text{ if, and only if, } y = \alpha_1 x, \alpha_1 \geq 0$$

□